WAQAS TARIQ TOOR, Ph.D.

BASIC INFO

Address	Chairman Office, Department of Electrical Engineering, UET Lahore (Narowal Campus), 10km Narowal-Muridke Road, Narowal
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EXPERIENCE

- **Academic Experience**
  - **Associate Professor** (May 2021 to date)
    - **University of Engineering and Technology, Lahore, Narowal Campus, Pakistan**
      Responsibilities: Looking after the matters of the department as Chairman, conducting accreditation visits, Teaching the graduate and undergraduate subjects of Electrical Engineering. Developing new labs and courses of engineering program. Carrying out research activities. Guiding students in final year projects and MS/PhD theses.
  - **Assistant Professor** (Jan 2018 to Apr 2021)
    - **Khwaja Fareed University of Engineering and Information Technology, RYK, Pakistan**
      Responsibilities: Teaching the graduate and undergraduate subjects of Electrical Engineering. Developing new labs and courses of engineering program. Carrying out research activities. Guiding students in final year projects and MS/PhD theses.
  - **Assistant Professor** (Oct 2009 to Mar 2014)
    - **University of Management & Technology Lahore, Pakistan**
      Responsibilities: Teaching the graduate and undergraduate subjects of Electrical Engineering. Developing new labs and courses of engineering program. Carrying out research activities. Guiding students in final year projects and MS Theses.
- **Non-Academic/Industrial Experience**
  - **Technical Expert** (Apr 2008 to Jun 2009)
    - **Alcatel-Lucent, Pakistan**
      Responsibilities: Operation and Management of Telecommunication sites while working in the field. Resolving issues of received signals, cards, and transmissions etc.

ACADEMIC MANAGEMENT & LEADERSHIP

- Chairman of the Department
- Curriculum Design and Review
- Convener, Final year design projects SOPs committee
- Convener, CQI committee of the department

COURSES TAUGHT

Undergraduate

- Wireless Communication
- Computer Networks
- Analog and Digital Communications
- Introduction to Machine Learning (Python)
- Computer Architecture
- Electronics I & II
- Circuits I & II
- Antenna and Wave Propagation
- Control Systems
- Electromagnetic Theory
- Digital Systems
- Digital Signal Processing
- Microprocessors Systems

Graduate

- Advanced Digital Communication
- Advanced Wireless Communication
- LTE Mobile Communication Systems (IoT)
- Digital Image Processing
- Linear Systems Theory
- Optimization Theory

EDUCATIONAL BACKGROUND

2014-18 **Ph.D.** (Electronics and Communication Engineering) **(QS Ranking:157)**
Hanyang University, Ansan, South Korea
Thesis: Multichannel Random Access for Wireless Networks with Application to Machine Type Communications
Advisor: Prof. Hu Jin

2009-11 **M.S.** (Electrical Engineering) **[Gold Medallist]**
University of Management & Technology, Lahore, Pakistan
Thesis: Neural Network Based Optimal Placement of Base Stations
Advisor: Prof. Abdul Aziz Bhatti

2003-07 **B.Sc.** (Electrical Engineering)
University of Engineering & Technology, Lahore, Pakistan
FYDP: Speed Control of Single Phase Induction Motor Using FPGA
Advisor: Prof. Zubair Ahmad Khan

PUBLICATIONS

- **Internet of Things (IoT)/ 5G Communication/Next Generation Networks**
  - **Transactions/Journals:**
    - Waqas Tariq Toor, Imran Javed**, "Capture Effect-based Optimal Access Class Barring for IoT Communications",** IEEE Access (Under Review) **[SCIE, Q1, IF: 3.9]**
    - Maira Alvi, Waqas Tariq Toor, Yawar Rehman, Imran Javed, Muhammad Idrees, Junaid Imtiaz, Muhammad Assam, Amira S. A. Aziz, E. A. Attia, **"Mini-slot Based Access Barring Scheme for IoT Networks"**, Human-Centric Computing and Information Sciences, Jul. 2023. **[SCIE, Q1, IF:6.558]**

- Ogobuchi Daniel Okey, Siti Sarah Maidin, Renata Lopes Rosa, Waqas Tariq Toor, Dick Carrillo Melgarejo, Lunchakorn Wuttisittikulkij, Muhammad Saadi, Demóstenes Zegarra Rodríguez, **"Quantum Key Distribution Protocol Selector Based on Machine Learning for Next-Generation Networks"**, Sustainaibility, vol. 14, no. 23, Nov. 2022. **[SCIE, Q2, IF:3.889]**
- Waqas Tariq Toor, Abdul Basit, Saqib Ali Khan, Naeem Maroof, **"Evolution of random access process: from legacy networks to 5G and beyond",** Transactions on Emerging Telecommunications Technology, vol. 33, no. 6, e3776, Aug. 2022. **[SCIE, Q1, IF:3.31]**
- Maira Alvi, Khamael M. Abualnaja, Waqas Tariq Toor, Muhammad Saadi, **"Performance analysis of access class barring for next generation IoT devices"**, Alexandria Engineering Journal, Vol. 60, Feb. 2021. **[SCIE, Q1, IF:6.626]**
- Jun Bae Seo, Waqas Tariq Toor, Hu Jin, **"Analysis of two-step random access procedure for cellular ultra-reliable low latency communications"**, IEEE Access, Vol. 9, Jan. 2021. **[SCIE, Q1, IF:3.9]**
- Waqas Tariq Toor, Maira Alvi, Mamta Agiwal, **"Combined access barring scheme for IoT devices using Bayesian estimation"**, Electronics, Vol. 9, Dec. 2020. **[SCIE, Q2, IF:2.9]**
- Mamta Agiwal, Ramesh Kumar, Waqas Tariq Toor, **"Two step random access latency improvement in congested beyond 5G networks"**, Electronics, Vol. 9, Nov. 2020. **[SCIE, Q2, IF:2.9]**
- Muhammad Saadi, Muhammad Talha Noor, Aadil Imran, Waqas Tariq Toor, Shahid Mumtaz, Lunchakorn Wuttisittikulkij, **"IoT enabled quality of experience measurement for next generation networks in smart cities"**, Sustainable Cities and Society, Vol. 60, Sep. 2020. **[SCIE, Q1, IF:10.69]**
- Hu Jin, Waqas Tariq Toor, Bang Chul Jung, Jun-Bae Seo, **"Recursive Pseudo-Bayesian access class barring for M2M communications in LTE systems"**, IEEE Transactions on Vehicular Technology, vol. 66, no. 9, pp. 8595-8599, Sep. 2017. **[SCIE, Q1, IF:7.11]**

- **Conferences**
  - Waqas Tariq Toor, Hu Jin, **"Combined access barring for energy and delay constrained machine type communications"**, IEEE International Conference on Technology Convergence (ICTC) 2018, Korea (South)
  - Waqas Tariq Toor, Hu Jin, **"Comparative study of access class barring and extended access barring for machine type communications"**, IEEE International Conference on Technology Convergence (ICTC) 2017, Korea (South)

- **Mobile Adhoc Networks (MANETs)**
  - **Transactions/Journals:**
    - Waqas Tariq Toor, Jun-Bae Seo, Hu Jin, **"Practical splitting algorithm for multi-channel slotted random access systems",** IEEE Transactions on Mobile Computing, Vol. 19, Dec. 2020. **[SCIE, Q1, IF:6.07]**

- Muhammad Saadi, Rabeel Manzoor, Waqas Tariq Toor, Sanika K Wijayasekara, Farhat Masood, Lunchakorn Wuttisittikulkij, **"A comparative study of customer preferences for telecommunication technologies in Pakistan"**, Engineering Journal, Vol. 24, No. 9, pp. 129-136, May 2020. **[ESCI, Q3]**
- Waqas Tariq Toor, Jun-Bae Seo, Hu Jin, "**Distributed transmission control in multichannel S-ALOHA for ad-hoc networks",** IEEE Communications Letters, vol. 21, no. 9, pp. 2093-2096, Sep. 2017. **[SCIE, Q1, IF:3.55]**

- **Conferences**
  - Waqas Tariq Toor, Jun-Bae Seo, and Hu Jin, **"Online control of random access with splitting,"** ACM MOBIHOC 2020, Oct. 11-14, 2020
  - Zhanyang Gao, Waqas Tariq Toor, Hu Jin, **"CDF based scheduling for uplink non-orthogonal multiple access"**, IEEE Vehicular Technology Conference (VTC) 2018, Portugal
  - Xuelian Wu, Waqas Tariq Toor, Hu Jin, **"On the CDF-based scheduling with non-uniform user distribution in multi-cell networks"**, IEEE International Conference on Technology Convergence (ICTC) 2017, Korea (South)
  - Waqas Tariq Toor, Abdul Aziz Bhatti, **"Neural Network Based Optimal Placement of Base Stations in Three-Dimensional Plane",** IEEE International Conference on Wireless Communication in Unusual and Confined Areas (ICWCUCA) 2012, France
  - Waqas Tariq Toor, Abdul Aziz Bhatti, Mashood Nasir, **"Comparison of Different Modulation Schemes for Signal Decoding in Analog Network Coding"**, IEEE 15th International Conference on Advanced Communication (ICACT) 2013, Korea (South)

- **Image Processing**
  - **Transactions/Journals:**
    - Waqas Tariq Toor, Farhan Naeem, Ihtisham ul Hassan, Muhammad Zubair Munir, **"Real Time Haze Removal for Camera Feed in Autonomous Vehicles"**, Sensors (Under Review) **[SCIE, Q2, IF:3.9]**
    - Abdul Basit, Waqas Tariq Toor, Naeem Maroof, Saqib Ali Khan, Sattam Al Otaibi, **"Reversible Encryption and Lossless Data Hiding for Medical Imaging Aiding Smart Health Care"**, Cluster Computing, Nov. 2022. **[SCIE, Q1, IF:4.4]**
    - Yawar Rehman, Hafsa Amanullah, DM Saqib Bhatti, Waqas Tariq Toor, Muhamad Ahmad, Manuel Mazzara, **"Detection of Small Size Traffic Signs Using Regressive Anchor Box Selection and DBL Layer Tweaking in YOLOv3"**, Applied Sciences, Vol. 11, Dec. 2021. **[SCIE, Q2, IF:2.7]**
    - Abdul Basit, Saqib Ali Khan, Waqas Tariq Toor, Naeem Maroof, Muhammad Saadi, Atif Khan **"Novel dissimilarity of activity biomarker and functional connectivity analysis for the epilepsy diagnosis",** Symmetry, vol. 11, no. 8, 979, Aug. 2019. **[SCIE, Q2, IF:2.7]**
  - **Conferences**

- Usman Ali, Waqas Tariq Toor, **"Mutually guided image dehazing",** International Conference on Emerging Technologies in Electronics, Computing, and Communications (ICETECC), Dec. 07-09, 2022

- **Miscellaneous**
  - **Transactions/Journals:**
    - Asif Hussain, Zafar Baig, Waqas Tariq Toor, Usman Ali, Muhammad Idrees, Tamara Al Shloul, Yazeed Yasin Ghadi, Hend Khalid Alkahtani, **"Wound Rotor Synchronous Motor as Promising Solution for Traction Applications"**, Electronics, Dec. 2022. **[SCIE, Q2, IF:2.9]**
  - **Conferences:**
    - Ateeqa Saeed, H. M. Salman Ajmal, M. Umair A. Khan, Waqas Tariq Toor, **"Enhancing Early Detection of Diabetes Using Machine Learning Algorithms",** International Conference on Innovative Engineering Sciences & Technological Research (ICIESTR), May 15-16 2024
    - M. Umair A. Khan, Qasim I. Khan, Waqas Tariq Toor, M. Rehan Chaudhry, H. M. Salman Ajmal, Sameen A. Malik, **"Design and Development of a Robotic Dental Surgical Manipulator",** International Conference on Innovative Engineering Sciences & Technological Research (ICIESTR), May 15-16 2024
    - Mashood Nasir, Abdul Aziz Bhatti, Waqas Tariq Toor, **"Model Formulation and Design of an Efficient Control Algorithm for Fuel Cell Power System",** IEEE 4th International Conference on Power Engineering Energy and Electrical Drives 2013, (POWERENG), Turkey

RESEARCH PROJECTS

- A study on mobile traffic capacity and optimal traffic distribution, Electronics and Telecommunications Research Institute (ETRI), 2016.6 - 2017.1.
- Development of random access techniques for low-delay clouldlet service, Gyeongsang National University, 2015.6 - 2015.12.
- Interference alignment in random access networks, Hanyang University, 2014.5 - 2015.4.
- A study on multi-user scheduling in multi-cell networks, Hanyang University, 2014.3 - 2015.2.
- Development of cloud radio access networks for extremely dense WLANs, National Research Foundation (NRF) of Korea, 2015.11 - 2018.10.
- Research and development of open 5G reference model, Giga Korea Foundation, 2016.8 - 2019.2.

MS THESES SUPERVISED

- Congestion Control in Internet of Things (IoT) Networks
- Reduction of Neural Network Memories Interconnection
- Transformer Less Design of Polymer Exchange Membrane Fuel Cell Based Uninterruptible Power Supply

RESEARCH INTERESTS

- Internet of Things (IoT)
- Smart Cities
- 5G, Next Generation Wireless Networks
- Scheduling
- Machine/Deep Learning
- Image Processing

AWARDS & HONORS

- Gold Medalist in MS Electrical Engineering
- National Scholarship for Pursuing Ph.D. Abroad, Higher Education Commission, Islamabad, Govt. of Pakistan

PROFESSIONAL MEMBERSHIPS

- Approved PhD Supervisor – Higher Education Commission of Pakistan
- Pakistan Engineering Council (ELECT/23350)

PROFESSIONAL TRAININGS

- Advanced OBE Training, UET Lahore, New Campus, Oct 2022
- Micro-teaching a tool for Quality Improvement, UET Lahore, Main Campus, Apr 2022
- Comprehensive Course Planning for Employability Skill Development, UET Lahore, New Campus, Feb 2022

REFERENCES

- Hu Jin, Associate Professor, School of Electrical Engineering, Hanyang University, South Korea
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  Tel: +82-31-400-5178
  Email: hjin@hanyang.ac.kr
- Qamar H. Abbasi (FRSA, FRET), Deputy Theme Lead, Quantum Technologies, University of Glasgow, United Kingdom
  Personal URL: https://www.gla.ac.uk/schools/engineering/staff/qammerabbasi/
  Tel: +447484650876
  Email: qammer.abbasi@glasgow.ac.uk
- Usman Ali, Assistant Professor, Department of Computer Science and Engineering, Sejong University, Seoul, South Korea
  Personal URL: https://sejong.elsevierpure.com/en/persons/usman-ali
  Tel: +82 10-6376-1485
  Email: usman.ali@sejong.ac.kr